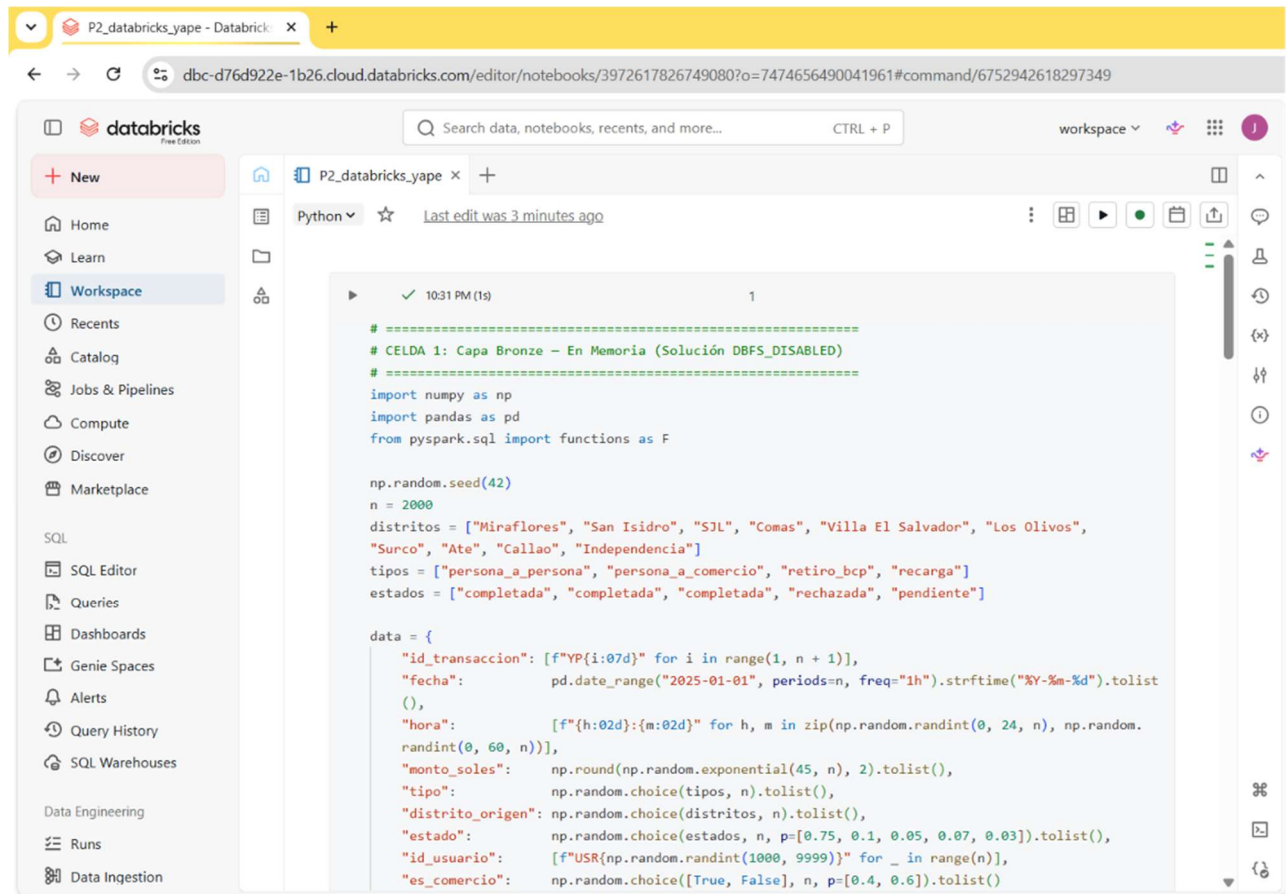


EP: BIG DATA

ALUMANA: ALANYA FLORES, JESICA ANDREA

DATABRICKS

Celda 1:



The screenshot shows the Databricks workspace interface. The top navigation bar includes a search bar and a workspace dropdown. The left sidebar contains a 'New' button and a list of navigation options: Home, Learn, Workspace (selected), Recents, Catalog, Jobs & Pipelines, Compute, Discover, and Marketplace. Below these are sections for SQL (SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, SQL Warehouses) and Data Engineering (Runs, Data Ingestion). The main area displays a Python notebook cell with the following code:

```
# =====  
# CELDA 1: Capa Bronze - En Memoria (Solución DBFS_DISABLED)  
# =====  
import numpy as np  
import pandas as pd  
from pyspark.sql import functions as F  
  
np.random.seed(42)  
n = 2000  
distritos = ["Miraflores", "San Isidro", "SOL", "Comas", "Villa El Salvador", "Los Olivos",  
"Surco", "Ate", "Callao", "Independencia"]  
tipos = ["persona_a_persona", "persona_a_comercio", "retiro_bcp", "recarga"]  
estados = ["completada", "completada", "completada", "rechazada", "pendiente"]  
  
data = {  
    "id_transaccion": [f"YP{i:07d}" for i in range(1, n + 1)],  
    "fecha": pd.date_range("2025-01-01", periods=n, freq="1h").strftime("%Y-%m-%d").tolist(),  
    "hora": [f"{h:02d}:{m:02d}" for h, m in zip(np.random.randint(0, 24, n), np.random.  
randint(0, 60, n))],  
    "monto_soles": np.round(np.random.exponential(45, n), 2).tolist(),  
    "tipo": np.random.choice(tipos, n).tolist(),  
    "distrito_origen": np.random.choice(distritos, n).tolist(),  
    "estado": np.random.choice(estados, n, p=[0.75, 0.1, 0.05, 0.07, 0.03]).tolist(),  
    "id_usuario": [f"USR{np.random.randint(1000, 9999)}" for _ in range(n)],  
    "es_comercio": np.random.choice([True, False], n, p=[0.4, 0.6]).tolist()
```

Celda 2:

dbc-d76d922e-1b26.cloud.databricks.com/editor/notebooks/3972617826749080?o=7474656490041961#command/6752942618297346

workspace

Python Last edit was 4 minutes ago

```

# =====
# CELDA 2: Capa Silver – Transformación desde Memoria
# =====
# En lugar de leer un archivo Parquet bloqueado, leemos la vista de la RAM
df_bronze_input = spark.table("v_bronze")

# Resolvemos los espacios vacíos del examen parcial
df_silver = df_bronze_input \
    .filter(F.col("estado") == "completada") \
    .withColumn("monto_pesos", F.round(F.col("monto_soles") * 1050, 2)) \
    .drop("es_comercio")

# Guardamos el resultado en una nueva vista temporal en memoria
df_silver.createOrReplaceTempView("v_silver")

print(f"✅ Capa Silver completada en memoria: {df_silver.count()} filas filtradas")
df_silver.show(5)

```

> See performance (3) ✓ Checked for improvements

> df_bronze_input: pyspark.sql.connect.dataframe.DataFrame = [id_transaccion: string, fecha: string ... 7 more fields]

> df_silver: pyspark.sql.connect.dataframe.DataFrame = [id_transaccion: string, fecha: string ... 7 more fields]

✅ Capa Silver completada en memoria: 1802 filas filtradas

id_transaccion	fecha	hora	monto_soles	tipo	distrito_origen	estado	id_usuario	mont o_pesos

Celda 3:

dbc-d76d922e-1b26.cloud.databricks.com/editor/notebooks/3972617826749080?o=7474656490041961#command/6752942618297346

workspace

Python Last edit was 4 minutes ago

```

# =====
# CELDA 3: Capa Gold – Agregaciones y Reporte de Negocio
# =====
# Leemos la información limpia desde la capa Silver en memoria
df_silver_input = spark.table("v_silver")

# Resolvemos los espacios vacíos del examen parcial:
# 1. Agrupamos por "distrito_origen"
# 2. Contamos el número de transacciones totales por distrito
# 3. Sumamos el monto total transaccionado en soles por distrito
# 4. Ordenamos de mayor a menor según el dinero movido
df_gold = df_silver_input \
    .groupBy("distrito_origen") \
    .agg(
        F.count("id_transaccion").alias("total_transacciones"),
        F.sum("monto_soles").alias("monto_total_soles")
    ) \
    .orderBy(F.col("monto_total_soles").desc())

# Mostramos el Top 10 de distritos final
print("👤 REPORTE EJECUTIVO YAPE: TOP DISTRITOS CON MAYOR MOVIMIENTO")
df_gold.show(10)

```

> See performance (1) ✓ Checked for improvements

> df_gold: pyspark.sql.connect.dataframe.DataFrame = [distrito_origen: string, total_transacciones: long ... 1 more field]

> df_silver_input: pyspark.sql.connect.dataframe.DataFrame = [id_transaccion: string, fecha: string ... 7 more fields]

👤 REPORTE EJECUTIVO YAPE: TOP DISTRITOS CON MAYOR MOVIMIENTO

Celda 4:

dbc-d76d922e-1b26.cloud.databricks.com/editor/notebooks/3972617826749080?o=7474656490041961#command/6752942618297351

Search data, notebooks, recents, and more... CTRL + P

workspace

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Spaces

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

P2_databricks_yape x +

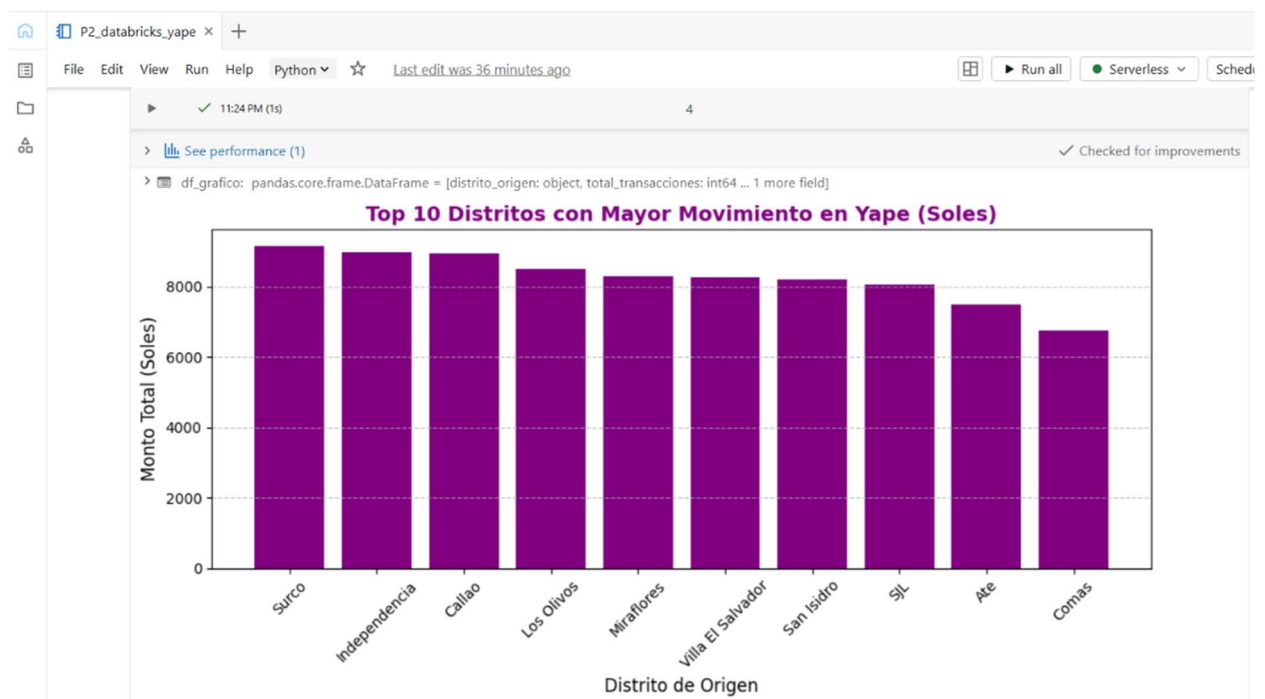
Python Last edit was 12 minutes ago

```
# =====  
# CELDA 4: Visualización – Gráfico de barras con Matplotlib  
# =====  
import matplotlib.pyplot as plt  
  
# 1. Pasamos el Top 10 de Spark a Pandas para poder graficarlo fácilmente  
df_grafico = spark.table("v_gold").limit(10).toPandas()  
  
# 2. Creamos la figura del gráfico  
plt.figure(figsize=(10, 5))  
plt.bar(df_grafico["distrito_origen"], df_grafico["monto_total_soles"], color="purple")  
  
# 3. Personalizamos el diseño con los colores de Yape  
plt.title("👤 Top 10 Distritos con Mayor Movimiento en Yape (Soles)", fontsize=14,  
fontweight="bold", color="purple")  
plt.xlabel("Distrito de Origen", fontsize=12)  
plt.ylabel("Monto Total (Soles)", fontsize=12)  
plt.xticks(rotation=45)  
plt.grid(axis="y", linestyle="--", alpha=0.7)  
  
# 4. Mostramos el gráfico en pantalla  
plt.tight_layout()  
plt.show()
```

> See performance (1) ✓ Checked for improvements

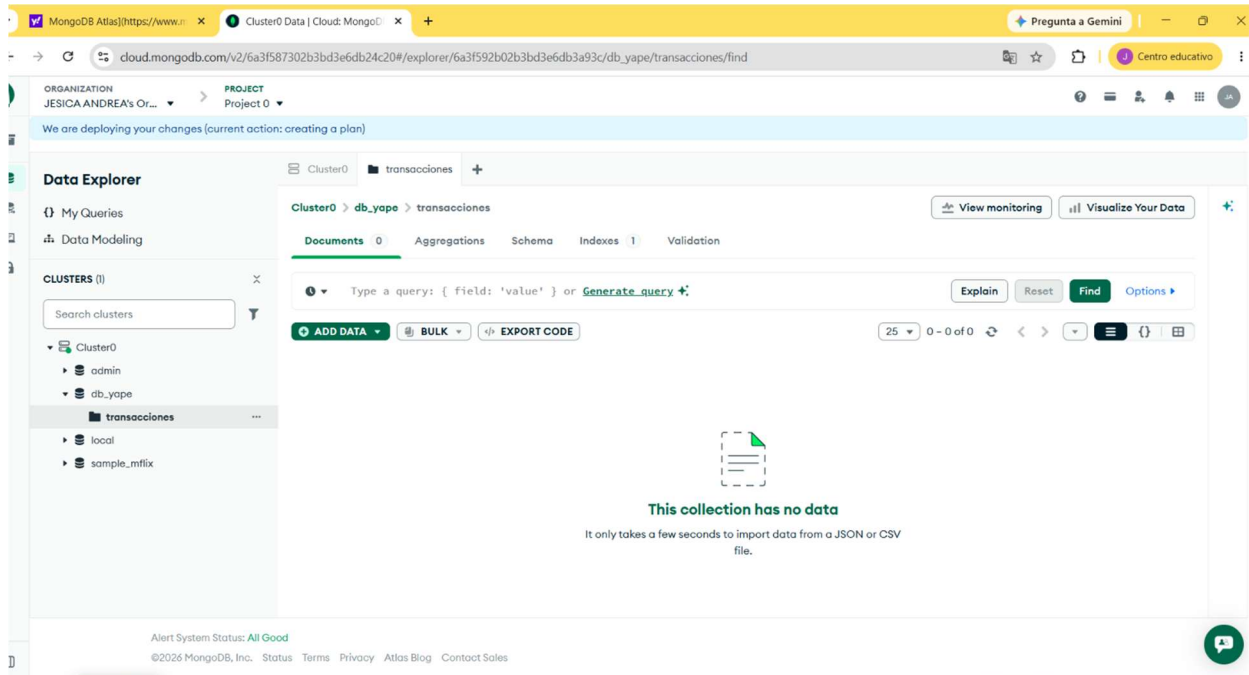
> df_grafico: pandas.core.frame.DataFrame = [distrito_origen: object, total_transacciones: int64 ... 1 more field]

/home/spark-e12c56d9-b763-4fc3-88da-60/.ipykernel/329/command-6752942618297351-5721883:21: UserWarning: G1

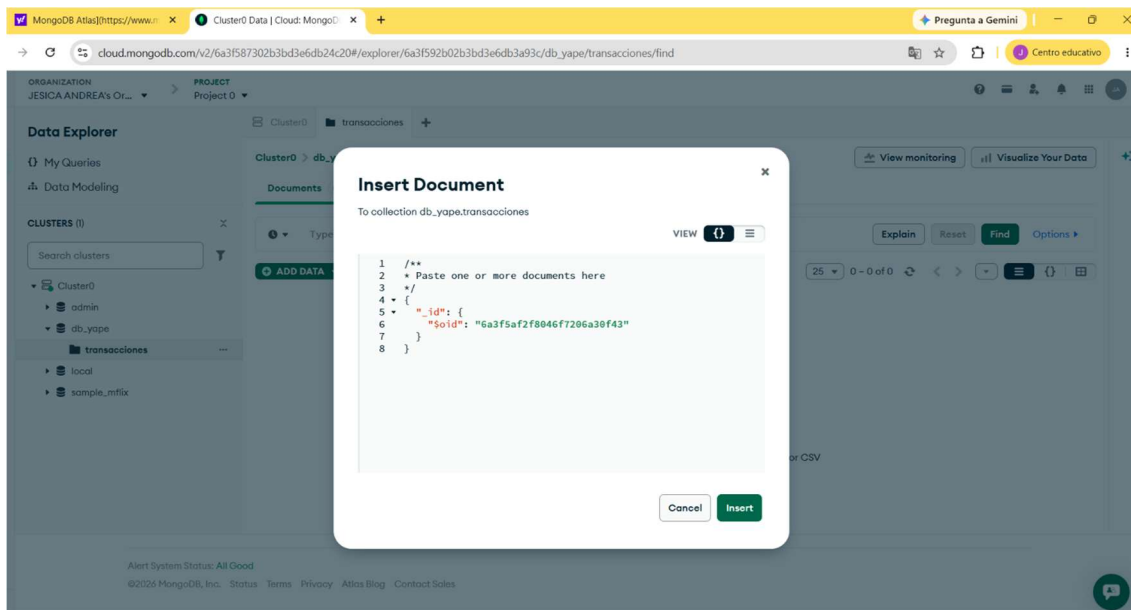


CREANDO CUENTA EN MONGODB

Creando carpeta de trabajo



Insertando código



Insert Document

To collection db_yape.transacciones

VIEW  

```
1 {
2   "id_transaccion": "YP0000001",
3   "fecha": {"$date": "2025-01-01T06:25:00Z"},
4   "monto_soles": {"$numberDecimal": "35.47"},
5   "tipo": "persona_a_persona",
6   "estado": "completada",
7   "usuario_origen": {
8     "id_usuario": "USR9055",
9     "celular": "987654321",
10    "distrito": "Surco"
11  },
12  "comercio_destino": {
13    "id_comercio": "COM4412",
14    "nombre_comercio": "Bodega Don Pepe",
15    "rubro": "Abarrotes"
16  }
17 }
```

Cancel

Insert

MongoDB Atlas[https://www.m... x Cluster0 Data | Cloud: MongoD... x +

cloud.mongodb.com/v2/6a3f587302b3bd3e6db24c20#/explorer/6a3f592b02b3bd3e6db3a93c/db_yape/transacciones/find

ORGANIZATION JESICA ANDREA's Or... PROJECT Project 0

Data Explorer

Cluster0 transacciones +

Cluster0 > db_yape > transacciones

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA BULK EXPORT CODE

```
_id: ObjectId('6a3f5b76f8046f7206a30f44')
id_transaccion: "YP0000001"
fecha: 2025-01-01T06:25:00.000+00:00
monto_soles: 35.47
tipo: "persona_a_persona"
estado: "completada"
usuario_origen: Object
comercio_destino: Object
```

Conectando MongoDB desde Colab

google colab - Yahoo Search T... x Untitled0.ipynb - Colab x + Pregunta a Gemini | - □ x

colab.research.google.com/drive/1uEcBUCEtI0Ktd5JOLk1Kg-EU6V8Np_A#scrollTo=wJqd... ☆ Centro educativo

Untitled0.ipynb ☆

Archivo Editar Ver Insertar Entorno de ejecución Herramientas Ayuda

Comandos + Código + Texto ▶ Ejecutar todas

RAM Disco

```
# =====
# INSTALACIÓN (en Google Colab: ejecutar con !)
# =====
# !pip install pymongo dnspython -q

from pymongo import MongoClient
import json

# REEMPLAZA con tu connection string de Atlas:
# Atlas → Connect → Drivers → Python → copia el string
CONNECTION_STRING = "mongodb+srv://jalanyaf_db_user:<db_password>@cluster0.fmgipwt.mongodb.net/"

client = MongoClient(CONNECTION_STRING)
db = client["yape_db"]
comerciantes = db["comerciantes"]
print("✅ Conectado a MongoDB Atlas")
print(f"DB: {db.name} | Colección: {comerciantes.name}")
```

... Conectado a MongoDB Atlas
DB: yape_db | Colección: comerciantes

Variables Terminal 0:44 Python 3

Filtros en MongoDB – “Comerciantes”

Consulta:

The screenshot shows the MongoDB Atlas Data Explorer interface. The left sidebar displays the 'Data Explorer' section with a tree view of clusters. The main panel shows the 'Cluster0 > db_yape > comerciantes' collection. A query is entered in the 'Find' tab: `{ "calificacion": { "$gt": 4.3 }, "yape_activo": true }`. The 'Sort' dropdown is set to `{ "calificacion": -1 }`. The 'Max Time MS' is set to 60000. The 'Collation' is `{ locale: 'simple' }`. The 'Skip' and 'Limit' values are 0. The 'Index Hint' is `{ field: -1 } or "indexName"`. The 'Find' button is highlighted. Below the query, a single document is displayed:

```
{
  "_id": ObjectId('6a3f6878fd6607ca9e056e9'),
  "ruc": "20512345678",
  "nombre_comercio": "Cevicheria El Muelle SAC",
  "tipo": "restaurante",
  "representante_legal": "Ana Flores Rojas",
  "distrito": "Miraflores"
}
```

At the bottom, the 'Alert System Status' is 'All Good'.

Resultado:

The screenshot shows the MongoDB Atlas Data Explorer interface with the same query as the previous screenshot. The 'Find' button is highlighted. Below the query, the results are displayed in a table with two documents:

nombre_comercio	tipo	calificacion
Cevicheria El Muelle SAC	restaurante	4.8
Farmacia San Pablo Express	Farmacia	4.5

The 'EXPORT CODE' button is highlighted. The 'INSIGHT' button is also visible. The 'Page' indicator shows 4.5.

Consulta:

ORGANIZATION JESICA ANDREA's Or... PROJECT Project 0

db_yape comerciantes +

Cluster0 > db_yape > comerciantes

Documents 5 Aggregations Schema Indexes 1 Validation

Tell us what to find (e.g. movies from 2000) or paste a query in another language (SQL, Java, etc.)

```
{
  "acepta_delivery": true,
  "departamento": "Lima",
  "monto_mensual_soles": { "$gt": 10000 }
}
```

Project { "nombre_comercio": 1, "monto_mensual_soles": 1, "distrito": 1, "_id": 0 }

Sort { field: -1 } or [['field', -1]]

Collation { locale: 'simple' }

Index Hint { field: -1 } or "indexName"

Resultado:

cloud.mongodb.com/v2/6a3f587302b3bd3e6db24c20#/explorer/6a3f592b02b3bd3e6db3a93c/db...

ORGANIZATION JESICA ANDREA's Or... PROJECT Project 0

db_yape comerciantes +

Cluster0 > db_yape > comerciantes View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 1 Validation

Tell us what to find (e.g. movies from 2000) or paste a query in another language (SQL, Java, etc.) Generate

```
{
  "acepta_delivery": true,
  "departamento": "Lima",
  "monto_mensual_soles": { "$gt": 10000 }
}
```

INSIGHT 25 1 - 3 of 3

nombre_comercio : "Farmacia San Pablo Express"
distrito : "Los Olivos"
monto_mensual_soles : 12000.00

nombre_comercio : "Distribuidora Norte SAC"
distrito : "Independencia"
monto_mensual_soles : 85000.00

Consulta:

ORGANIZATION JESICA ANDREA's Or... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - db_yape
 - comerciantes
 - transacciones
 - local
 - sample_mflix

Cluster0 > db_yape > comerciantes

Documents 5 Aggregations Schema Indexes 1 Validation

Tell us what to find (e.g. movies from 2000) or paste a query in another language (SQL, Java, etc.)

`{ "tipo": { "$in": ["bodega", "farmacia"] } }` Explain Reset Find Options

Project { "nombre_comercio": 1, "tipo": 1, "_id": 0 }

Sort { field: -1 } or [['field', -1]] Max Time MS 60000

Collation { locale: 'simple' } Skip 0 Limit 0

Index Hint { field: -1 } or "indexName"

25 1 - 3 of 3

Resultado:

ORGANIZATION JESICA ANDREA's Or... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - db_yape
 - comerciantes
 - transacciones
 - local
 - sample_mflix

Cluster0 > db_yape > comerciantes

Documents 5 Aggregations Schema Indexes 1 Validation

Tell us what to find (e.g. movies from 2000) or paste a query in another language (SQL, Java, etc.)

`{ "tipo": { "$in": ["bodega", "farmacia"] } }` Explain Reset Find Options

25 1 - 2 of 2

nombre_comercio: "Bodega La Esquina de Don Mario"
tipo: "bodega"

nombre_comercio: "Farmacia San Pablo Express"
tipo: "farmacia"